

- **Lab Title:** Lab 3 Data Carving with XXD, Binwalk and Scalpel
- **Assignment Title:** Data Carving with XXD, Binwalk and Scalpel
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- **Course Name:** SBT-DF202 Computer and Digital Forensics
- **Instructor Name:** Aminu Idris
- **Date of Submission:** 4 Sep, 2026
- **Version:** Version 2.2.2

Platform: Kali Linux or Ubuntu Forensic Workstation / VM

Lab Overview

By completing this lab, students will demonstrate their ability to explain why data carving is required when file-system metadata is missing or unreliable; use xxd to examine binary file content and identify headers and footers; recognise JPEG SOI and EOI signatures; reconstruct binary content from a hexadecimal dump; use The Sleuth Kit to examine inode, block and sector-level data; identify and extract embedded content using Binwalk; recover deleted files from a forensic USB image using Scalpel; verify recovered evidence using cryptographic hashes; and prepare a professional forensic report containing commands, screenshots, findings and limitations.

Introduction

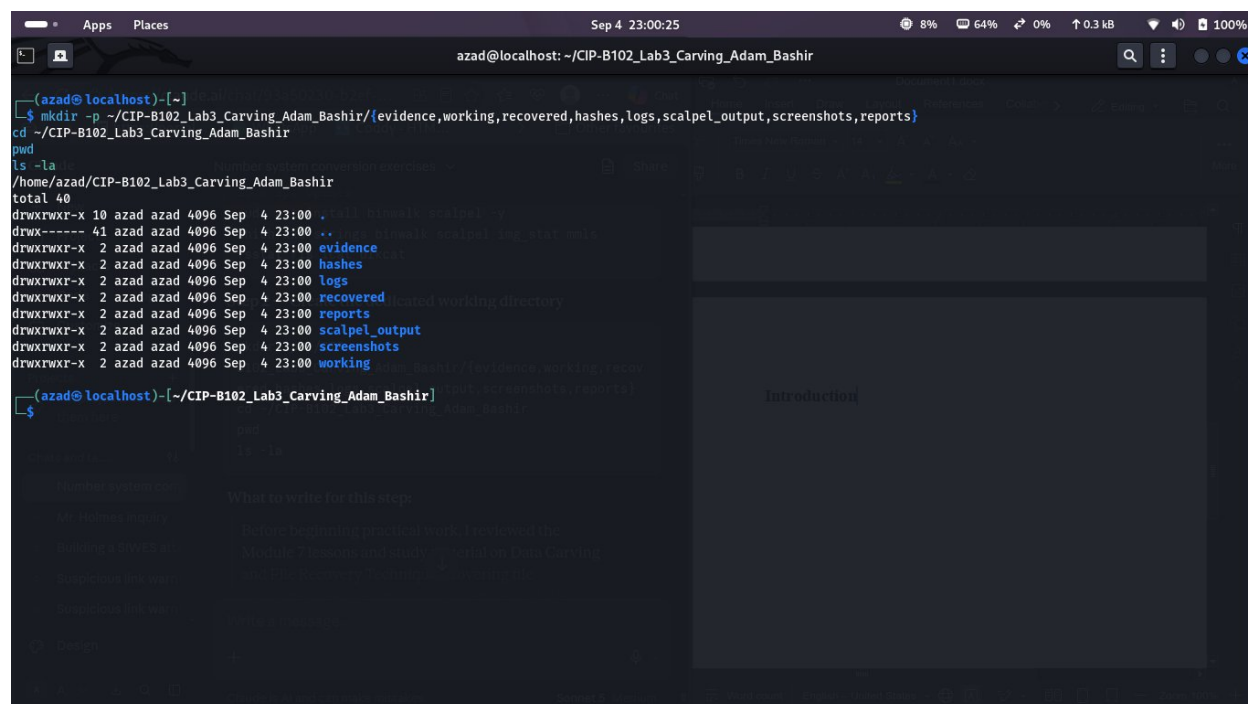
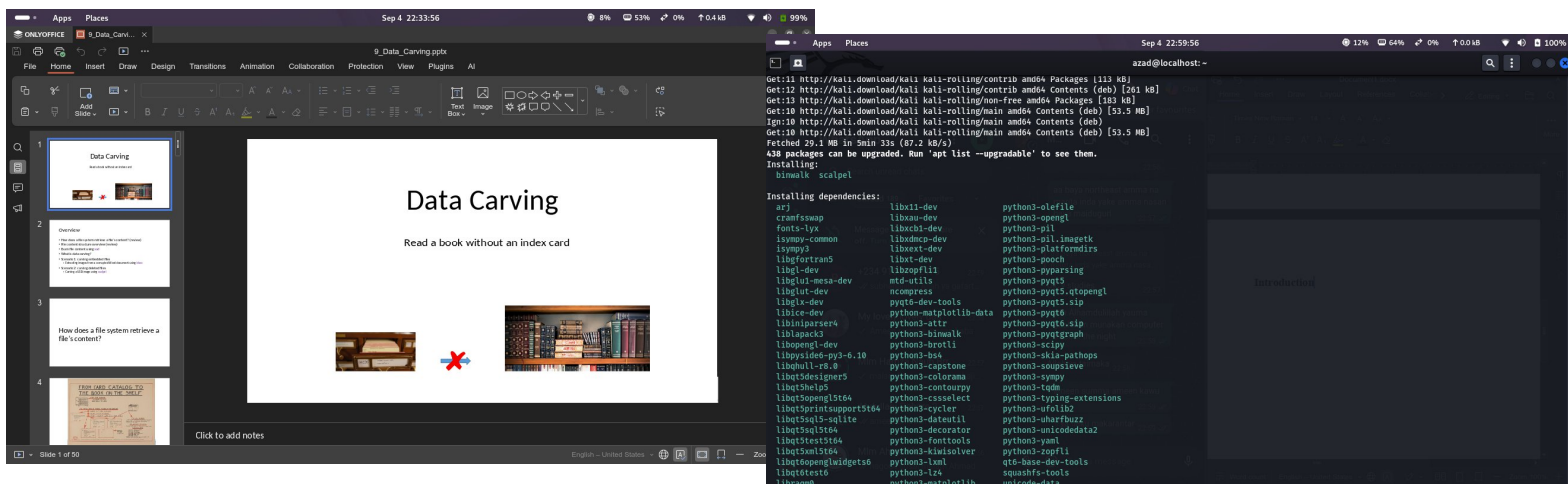
This report documents a data carving and file recovery exercise conducted as a junior digital forensic analyst practical. The scenario required examining raw binary file content, identifying file signatures at the byte level, recovering embedded and deleted content without relying on filesystem metadata, and carving deleted files directly from a forensic USB image skills that become essential when a filesystem's own metadata is missing, damaged, or deliberately unreliable.

Before beginning any hands-on work, I reviewed the Module 7 lessons and study material on Data Carving and File Recovery Techniques, covering file signatures, hexadecimal analysis, embedded-file recovery, and carving concepts. Three pieces of authorized training evidence were used throughout: J_ub_law.jpg (a JPEG for hex/signature analysis), Ch01InChap01.dd (a FAT12 floppy image reused from an earlier lab), and 120M.7z (a compressed FTK-acquired USB image containing dozens of deliberately deleted files of many types). All evidence was hashed (MD5 and SHA-256) before any analysis began, and the original files were never modified all carving, extraction, and reconstruction was performed into separate working and output folders.

The investigation progressed through byte-level signature analysis with xxd (Part A), metadata string extraction (Part B), direct data-unit examination with The Sleuth Kit (Part C), embedded-content discovery with Binwalk (Part D), USB image preparation and partition confirmation (Part E), and finally signature-based file carving with Scalpel (Part F) each step building toward the same underlying principle: that a file's true identity is defined by its raw byte content and structure, not by its filename, extension, or filesystem metadata, all of which can be missing, altered, or deliberately misleading.

Step 1 Confirm required tools are installed, Create the dedicated working directory

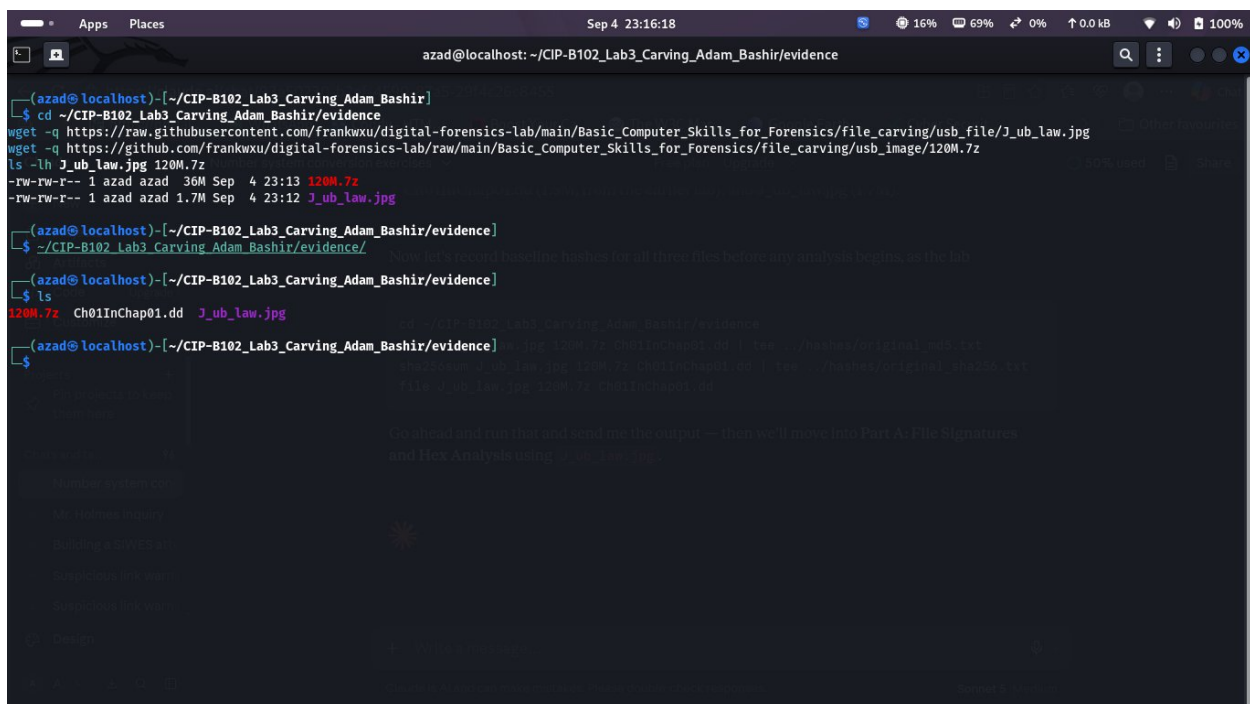
Before beginning practical work, I reviewed the Module 7 lessons and study material on Data Carving and File Recovery Techniques, covering file signatures, hexadecimal analysis, embedded-file recovery, and carving concepts, as required by the lab's instructor briefing. I then confirmed all required tools were available xxd, strings, binwalk, scalpel, and the Sleuth Kit tools (img_stat, mmls, fsstat, fls, icat, blkcat) installing any that were missing. I created a dedicated working directory with separate subfolders for evidence, working copies, recovered/carved files, hashes, logs, Scalpel output, screenshots, and reports, keeping this lab's evidence completely separate from my earlier Autopsy/Sleuth Kit lab.



Step 2 Get the three evidence files into evidence/

I downloaded the two additional required training evidence files from the publicly available, DOJ-funded academic teaching repository [frankwxu/digital-forensics-lab](https://github.com/frankwxu/digital-forensics-lab) on GitHub, which hosts the exact evidence files referenced by this lab's Module 7 material. `J_ub_law.jpg` (1.7M) is a JPEG sample intended for header/footer signature and hex analysis, and `120M.7z` (36M, a compressed archive containing a 120MB USB image) is intended for Scalpel-based file carving. I also copied `Ch01InChap01.dd` from my earlier Autopsy/Sleuth Kit lab into this lab's dedicated evidence folder, keeping the two labs' working directories fully separate despite reusing the same source image.

`ls -lh` confirmed all three evidence files are present: `120M.7z` (36M), `Ch01InChap01.dd` (1.5M, from the earlier lab), and `J_ub_law.jpg` (1.7M).

A terminal window titled 'azad@localhost: ~/CIP-B102_Lab3_Carving_Adam_Bashir/evidence' shows the following commands and output:
1. `cd ~/CIP-B102_Lab3_Carving_Adam_Bashir/evidence`
2. `wget -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/file_carving/usb_file/J_ub_law.jpg`
3. `wget -q https://github.com/frankwxu/digital-forensics-lab/raw/main/Basic_Computer_Skills_for_Forensics/file_carving/usb_image/120M.7z`
4. `ls -lh J_ub_law.jpg 120M.7z`
Output for step 4:
-rw-rw-r-- 1 azad azad 36M Sep 4 23:13 120M.7z
-rw-rw-r-- 1 azad azad 1.7M Sep 4 23:12 J_ub_law.jpg
5. `ls`
Output for step 5:
120M.7z Ch01InChap01.dd J_ub_law.jpg

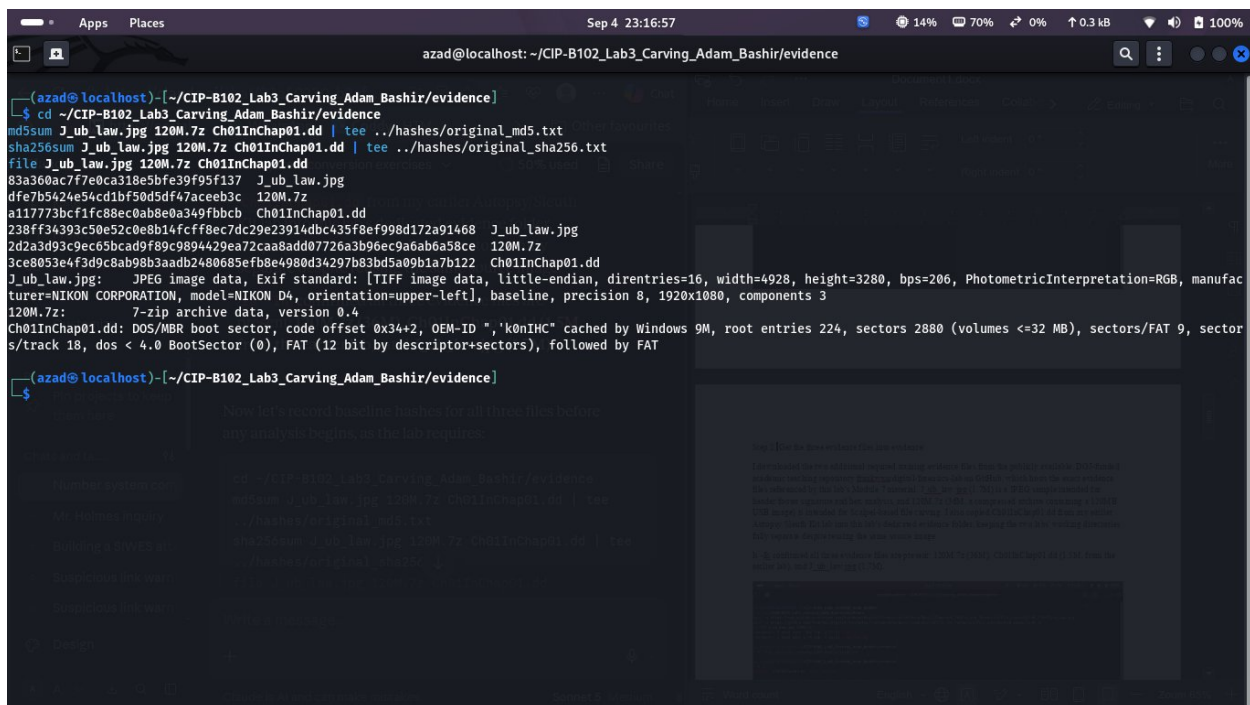
I recorded MD5 and SHA-256 hashes for all three files before any analysis begins, as the lab

File	MD5	SHA-256
J_ub_law.jpg	83a360ac7f7e0ca318e5bfe39f95f137	238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468
120M.7z	dfe7b5424e54cd1bf50d5df47aceeb3c	2d2a3d93c9ec65bcad9f89c9894429ea72caa8add07726a3b96ec9a6ab6a58ce
Ch01InChap01.dd	a117773bcf1fc88ec0ab8e0a349fbbcb	3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122

Note: Ch01InChap01.dd's hashes here exactly match the hashes I recorded for the same file in my previous Autopsy lab confirming this is genuinely the same, unaltered evidence file being reused across both labs.

file identified each evidence type:

- J_ub_law.jpg is a genuine JPEG with embedded EXIF metadata, revealing it was captured on a NIKON D4 camera, with a sensor resolution of 4928×3280, though the actual embedded JPEG image itself is baseline 1920×1080. This EXIF preview is useful context heading into Part B's metadata search.
- 120M.7z is confirmed as a valid 7-Zip archive (version 0.4 format).
- Ch01InChap01.dd is confirmed as the same FAT12 floppy image identified previously, further detailing 224 root entries, 2880 sectors, and 9 sectors/FAT



```
(azad@localhost)-[~/CIP-B102_Lab3_Carving_Adam_Bashir/evidence]
$ cd ~/CIP-B102_Lab3_Carving_Adam_Bashir/evidence
md5sum J_ub_law.jpg 120M.7z Ch01InChap01.dd | tee ../hashes/original_md5.txt
sha256sum J_ub_law.jpg 120M.7z Ch01InChap01.dd | tee ../hashes/original_sha256.txt
file J_ub_law.jpg 120M.7z Ch01InChap01.dd
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_law.jpg
dfe7b5424e54cd1bf50d5df47aceeb3c 120M.7z
a117773bcf1fc88ec0ab8e0a349fbbcb Ch01InChap01.dd
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 J_ub_law.jpg
2d2a3d93c9ec65bcad9f89c9894429ea72caa8add07726a3b96ec9a6ab6a58ce 120M.7z
3ce8053e4f3d9c8ab98b3aadbb2480685efb8e4980d34297b83bd5a09b1a7b122 Ch01InChap01.dd
J_ub_law.jpg: JPEG image data, Exif standard: [TIFF image data, little-endian, direntries=16, width=4928, height=3280, bps=206, PhotometricInterpretation=RGB, manufac
turer=NIKON CORPORATION, model=NIKON D4, orientation=upper-left], baseline, precision 8, 1920x1080, components 3
120M.7z: 7-zip archive data, version 0.4
Ch01InChap01.dd: DOS/MBR boot sector, code offset 0x34+2, OEM-ID "", "k0nIHC" cached by Windows 9M, root entries 224, sectors 2880 (volumes <=32 MB), sectors/FAT 9, sector
s/track 18, dos < 4.0 BootSector (0), FAT (12 bit by descriptor+sectors), followed by FAT
```

(azad@localhost)-[~/CIP-B102_Lab3_Carving_Adam_Bashir/evidence]

```
$
```

Note: Let's record baseline hashes for all three files before any analysis begins on them requires

```
md5sum J_ub_law.jpg 120M.7z Ch01InChap01.dd | tee ../hashes/original_md5.txt
sha256sum J_ub_law.jpg 120M.7z Ch01InChap01.dd | tee ../hashes/original_sha256.txt
```

File analysis results:

- J_ub_law.jpg: JPEG image data, Exif standard: [TIFF image data, little-endian, direntries=16, width=4928, height=3280, bps=206, PhotometricInterpretation=RGB, manufacturer=NIKON CORPORATION, model=NIKON D4, orientation=upper-left], baseline, precision 8, 1920x1080, components 3
- 120M.7z: 7-zip archive data, version 0.4
- Ch01InChap01.dd: DOS/MBR boot sector, code offset 0x34+2, OEM-ID "", "k0nIHC" cached by Windows 9M, root entries 224, sectors 2880 (volumes <=32 MB), sectors/FAT 9, sectors/track 18, dos < 4.0 BootSector (0), FAT (12 bit by descriptor+sectors), followed by FAT

Part A — File Signatures and Hex Analysis

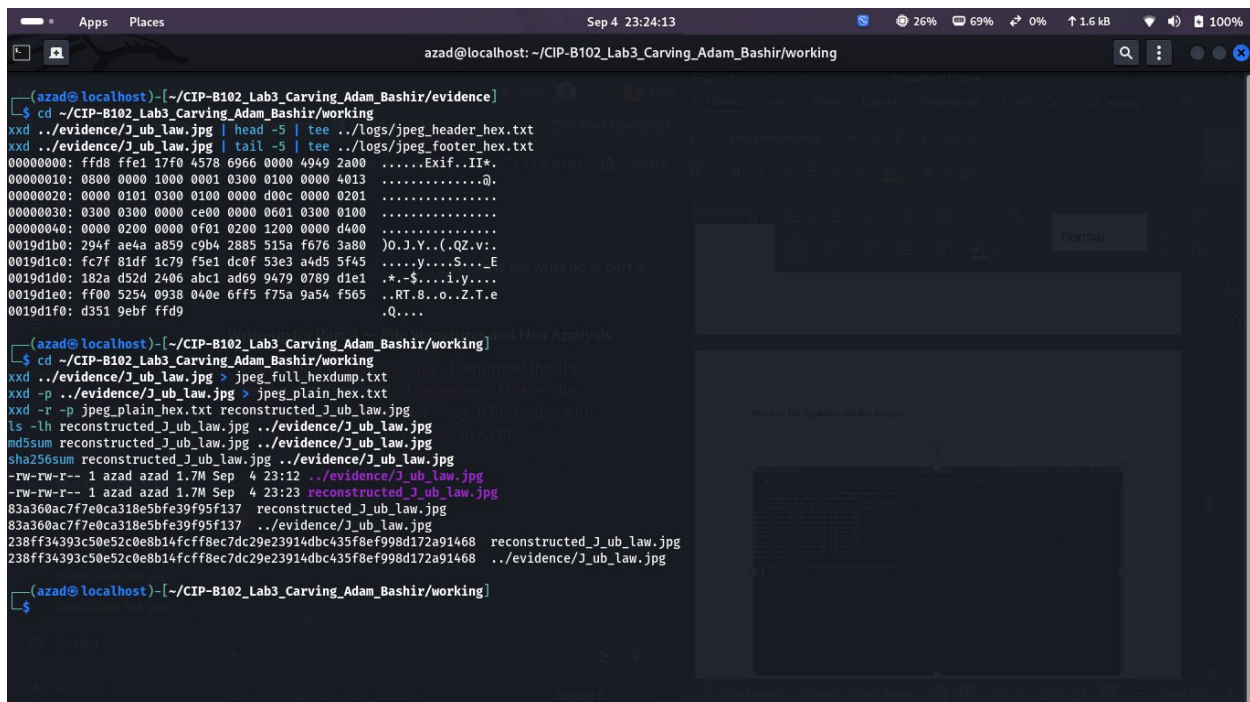
Using xxd on J_ub_law.jpg, I confirmed the JPEG header signature at offset 00000000: ffd8 the Start of Image (SOI) marker every JPEG begins with, immediately followed by ffe1, an APP1 marker indicating embedded Exif metadata. At the very end of the file (offset 0019d1f0), the last two bytes are ffd9 the End of Image (EOI) marker, confirming the file terminates correctly with a valid JPEG footer.

I then created a full annotated hex dump (xxd) and a plain hex dump (xxd -p) of the file. Using the plain hex dump, I reconstructed the image from scratch with xxd -r -p, converting the hexadecimal text back into raw binary bytes.

ls -lh confirmed the reconstructed file matched the original size exactly: 1.7M for both. md5sum and sha256sum confirmed both hashes were identical between the original and the reconstructed copy:

- MD5: 83a360ac7f7e0ca318e5bfe39f95f137 (both files)
- SHA256: 238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 (both files)

This proves the hex-dump-and-reconstruct round trip is lossless converting a file to hex text and back reproduces it byte-for-byte identically. This is the same underlying principle used in data carving: if a file's raw bytes can be correctly identified and extracted (by their header/footer signatures), the file can be perfectly reconstructed even without relying on filesystem metadata.



```
azad@localhost: ~/CIP-B102_Lab3_Carving_Adam_Bashir/working
$ cd ~/CIP-B102_Lab3_Carving_Adam_Bashir/evidence
$ xxd ../evidence/J_ub_law.jpg | head -5 | tee ../logs/jpeg_header_hex.txt
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00 .....Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013 .....a.
00000020: 0000 0101 0300 0100 0000 d00c 0000 0201 .....
00000030: 0300 0300 0000 ce00 0000 0601 0300 0100 .....
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400 .....
0019d1b0: 294f ae4a a859 c9b4 2885 515a f676 3a80 )0.J.Y..(.QZ.v:
0019d1c0: fc7f 81df 1c79 f5e1 dc0f 53e3 a4d5 5f45 .....y....S...E
0019d1d0: 182a d52d 2406 abc1 ad09 9479 0789 d1e1 *.~$.~.i.y....
0019d1e0: ff00 5254 0938 040e 6ff5 f75a 9a54 f565 ..RT.8..o..Z.T.e
0019d1f0: d351 9ebf ffd9 .....Q....

$ xxd -p ../evidence/J_ub_law.jpg > jpeg_full_hexdump.txt
$ xxd -p ../evidence/J_ub_law.jpg > jpeg_plain_hex.txt
$ xxd -r -p jpeg_plain_hex.txt reconstructed_J_ub_law.jpg
$ ls -lh reconstructed_J_ub_law.jpg ../evidence/J_ub_law.jpg
-rw-rw-r-- 1 azad azad 1.7M Sep  4 23:12 ../evidence/J_ub_law.jpg
-rw-rw-r-- 1 azad azad 1.7M Sep  4 23:23 reconstructed_J_ub_law.jpg
$ md5sum reconstructed_J_ub_law.jpg ../evidence/J_ub_law.jpg
83a360ac7f7e0ca318e5bfe39f95f137 reconstructed_J_ub_law.jpg
83a360ac7f7e0ca318e5bfe39f95f137 ../evidence/J_ub_law.jpg
$ sha256sum reconstructed_J_ub_law.jpg ../evidence/J_ub_law.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 reconstructed_J_ub_law.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 ../evidence/J_ub_law.jpg

$
```

Part B - Examine Embedded Metadata-Like Content

Using strings combined with a case-insensitive keyword filter, I found substantial embedded metadata inside J_ub_law.jpg, recorded exactly as found (no items guessed or assumed):

Metadata Item	Value Found
Camera model	NIKON D4 (manufacturer: NIKON CORPORATION)
Lens	17.0–35.0 mm f/2.8
Camera serial number	2027750
Editing software	Adobe Photoshop 21.1 (Macintosh) , with earlier edits via Adobe Photoshop CS6 and Camera Raw 8.3
Original creation date	2013-10-08 , 18:56:21
Last metadata/save date	2020-08-20 , 10:20:05 (–04:00)
Author / rights holder	HOWARD KORN
Original raw filename	UB_10-8-13_cam2_0117.jpg
Subject/keyword tags	"2013", "campus", "Merrick School of Business"
Hierarchical subject	Schools Merrick School of Business
Color profile copyright	Copyright 1999 Adobe Systems Incorporated (generic Adobe RGB (1998) ICC profile string, not the photo's own copyright)
GPS/location coordinates	Not found

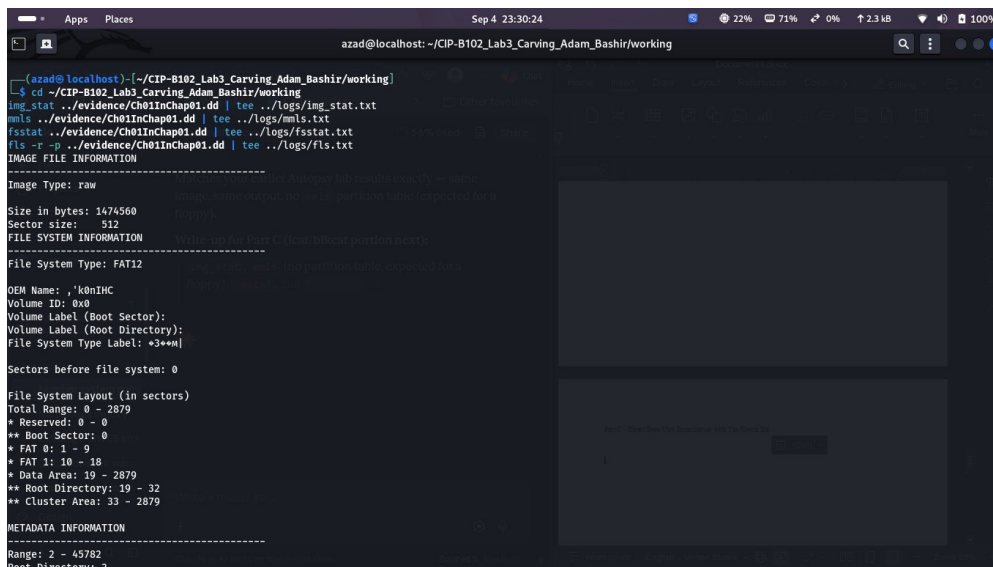
This is a rich metadata trail: the image was originally captured on **8 October 2013** with a Nikon D4, then went through multiple Photoshop editing sessions (visible in the full XMP edit history, spanning January 2014 through August 2020) before being finalized. The subject tags and the original raw filename (UB_...cam2...) strongly suggest this was a **University of Baltimore campus photo**, credited to Howard Korn. No GPS coordinates were present in the extracted strings, so that item is recorded as **not found**, per the lab's instruction not to guess.

```
Apps Places Sep 24 23:27:16 azad@localhost: ~/CIP-B102_Lab3_Carving_Adam_Bashir/working

(azad@localhost)~/CIP-B102_Lab3_Carving_Adam_Bashir/working
$ cd ~/CIP-B102_Lab3_Carving_Adam_Bashir/working
strings ../evidence/J_ub_law.jpg | grep -i -E "nikon|adobe|photoshop|canon|software|camera|2019|2020|2021|2022|2023" | tee ../logs/jpeg_strings_metadata.txt
NIKON CORPORATION
NIKON D4
Adobe Photoshop 21.1 (Macintosh)
2020:08:20 10:20:05
Adobe_CM
Adobe
"Photoshop 3.0
Adobe_CM
Adobe
http://ns.adobe.com/xap/1.0/
" id="W5M0MpcCehiHzrsZNTczkc9d"?> <x:xmpmeta xmlns:x="adobe:meta/" x:xmptk="Adobe XMP Core 6.0-c002 79.164352, 2020/01/30-15:50:38"
//www.w3.org/1999/02/22-rdf-syntax-ns#"> <rdf:Description rdf:about="" xmlns:aux="http://ns.adobe.com/exif/1.0/aux/" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:ph
otshop="http://ns.adobe.com/photoshop/1.0/" xmlns:xmp="http://ns.adobe.com/xap/1.0/" xmlns:xmpMM="http://ns.adobe.com/xap/1.0/mm/" xmlns:stRef="http://ns.adobe.com/xap/
1.0/stType/ResourceRef#" xmlns:stEvt="http://ns.adobe.com/xap/1.0/stType/ResourceEvent#" xmlns:crs="http://ns.adobe.com/camera-raw-settings/1.0/" xmlns:lr="http://ns.adobe
.com/lightroom/1.0/" aux:ApproximateFocusDistance="4294967295/1" aux:ImageNumber="115422" aux:Lens="17.0-35.0 mm f/2.8" aux:LensID="99" aux:LensInfo="170/10 350/10 28/10
28/10" aux:SerialNumber="2027750" dc:format="image/jpeg" photoshop:DateCreated="2013-10-08T18:56:21" photoshop:LegacyIPTCDigest="5E379AE71DB03222FA2DE651E47F460" photoshop:Colo
rMode="3" photoshop:ICPProfiles="Adobe RGB (1998)" xmp:CreateDate="2013-10-08T18:56:21" xmp:CreatorTool="NIKON D4 Ver.1.01" xmp:Label="Approved" xmp:MetadataDate="2020-0
8-20T10:20:05-04:00" xmp:ModifyDate="2020-08-20T10:20:05-04:00" xmp:Rating="0" xmpMM:DocumentID="adobe:docid:photoshop:307cd9db-3a87-6148-9874-c6730fd42952" xmpMM:Instan
ceID="xmp.iid:9e7f6b27-04e9-43e7-ae79-c3f2fb9c2ac9" xmpMM:OriginalDocumentID="B04B02E367C62737F8E78649729D6364" crs:AlreadyApplied="True" crs:AutoLateralCA="0" crs:AutoW
hiteVersion="134348800" crs:Blacks2012="0" crs:BlueHue="0" crs:BlueSaturation="0" crs:CameraProfile="Embedded" crs:CameraProfileDigest="54650A341B5B5CCAE8442D0B43A92BCE"
crs:Clarity2012="0" crs:ColorNoiseReduction="0" crs:Contrast2012="0" crs:ConvertToGrayscale="False" crs:DefringeGreenAmount="0" crs:DefringeGreenHueHi="60" crs:Defringe
GreenHueLo="40" crs:DefringePurpleAmount="0" crs:DefringePurpleHueHi="70" crs:DefringePurpleHueLo="30" crs:Exposure2012="0.00" crs:GrainAmount="0" crs:GreenHue="0" crs:G
reenSaturation="0" crs:HasCrop="False" crs:HasSettings="True" crs:Highlights2012="0" crs:HueAdjustmentAqua="0" crs:HueAdjustmentBlue="0" crs:HueAdjustmentGreen="0" crs:H
ueAdjustmentMagenta="0" crs:HueAdjustmentOrange="0" crs:HueAdjustmentPurple="0" crs:HueAdjustmentRed="0" crs:HueAdjustmentYellow="0" crs:IncrementalTemperature="0" crs:I
ncrementalTint="0" crs:LensManualDistortionAmount="0" crs:LensProfileEnable="0" crs:LensProfileSetup="LensDefaults" crs:LuminanceAdjustmentAqua="0" crs:LuminanceAdjustmen
tBlue="0" crs:LuminanceAdjustmentGreen="0" crs:LuminanceAdjustmentMagenta="0" crs:LuminanceAdjustmentOrange="0" crs:LuminanceAdjustmentPurple="0" crs:LuminanceAdjustmen
tRed="0" crs:LuminanceAdjustmentYellow="0" crs:LuminanceSmoothing="0" crs:ParametricDarks="0" crs:ParametricHighlightSplit="75" crs:ParametricHighlights="0" crs:Parametr
icLights="0" crs:ParametricMidtoneSplit="50" crs:ParametricShadowSplit="25" crs:ParametricShadows="0" crs:PerspectiveAspect="0" crs:PerspectiveHorizontal="0" crs:Perspec
tiveRotate="0.0" crs:PerspectiveScale="100" crs:PerspectiveUpright="0" crs:PerspectiveVertical="0" crs:PostCropVignetteAmount="0" crs:ProcessVersion="6.7" crs:RawFileNam
e="UB_10-8-13_cam2_0117.jpg" crs:RedHue="0" crs:RedSaturation="0" crs:Saturation="0" crs:SaturationAdjustmentAqua="0" crs:SaturationAdjustmentBlue="0" crs:SaturationAdju
dgmentGreen="0" crs:SaturationAdjustmentMagenta="0" crs:SaturationAdjustmentOrange="0" crs:SaturationAdjustmentPurple="0" crs:SaturationAdjustmentRed="0" crs:SaturationA
djustmentYellow="0" crs:ShadowTint="0" crs:Shadows2012="0" crs:SharpenDetail="25" crs:SharpenEdgeMasking="0" crs:SharpenRadius="+1.0" crs:Sharpness="0" crs:SplitToningBa
lance="0" crs:SplitToningHighlightHue="0" crs:SplitToningHighlightSaturation="0" crs:SplitToningShadowHue="0" crs:SplitToningShadowSaturation="0" crs:ToneCurveName2012="
Linear" crs:Version="8.3" crs:Vibrance="0" crs:VignetteAmount="0" crs:WhiteBalance="As Shot" crs:Whites2012="0"> <dc:subject> <rdf:Bag> <rdf:li>2013</rdf:li> <rdf:li>cam
pus</rdf:li> <rdf:li>Merrick School of Business</rdf:li> </rdf:Bag> </dc:subject> <dc:rights> <rdf:Alt> <rdf:li xml:lang="x-default">HOWARD KORN</rdf:li> </rdf:Alt> </dc
```

Part C - Direct Data-Unit Examination with The Sleuth Kit

img_stat, mmls (no partition table, expected for a floppy), fsstat, and fls -r -p all reproduced identical results to my earlier Sleuth Kit analysis of this same image: FAT12, 1,474,560 bytes, 2880 sectors, with 6 total files (2 live: Client Info.mdb, Income.xls; 4 deleted: Billing Letter.doc, confirmation.txt, letter1.txt, Regrets.doc). This confirms the image is unmodified across labs, consistent with my earlier baseline hash match.



```
azad@localhost: ~/CIP-B102_Lab3_Carving_Adam_Bashir/working
$ cd ~/CIP-B102_Lab3_Carving_Adam_Bashir/working
$ img_stat ../evidence/Ch01InChap01.dd | tee ../logs/img_stat.txt
$ mmls ../evidence/Ch01InChap01.dd | tee ../logs/mmls.txt
$ fsstat ../evidence/Ch01InChap01.dd | tee ../logs/fsstat.txt
$ fls -r -p ../evidence/Ch01InChap01.dd | tee ../logs/fls.txt

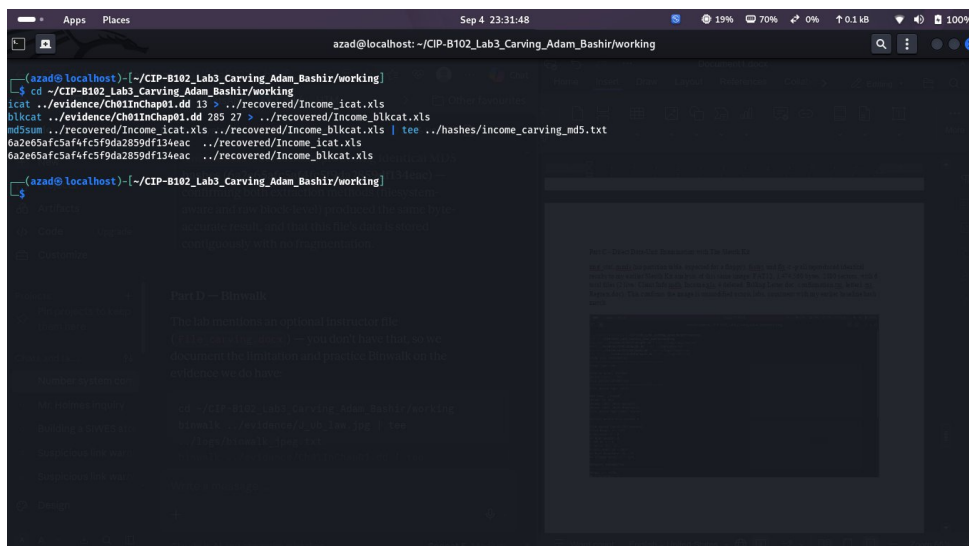
IMAGE FILE INFORMATION
-----
Image Type: raw
Size in bytes: 1474560
Sector size: 512
FILE SYSTEM INFORMATION
-----
File System Type: FAT12
ODM Name: 'k0n1MC'
Volume ID: 0x0
Volume Label (Root Sector):
Volume Label (Root Directory):
File System Type Label: *3+*m|

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 2879
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 9
* FAT 1: 10 - 18
* Data Area: 19 - 2879
** Root Directory: 19 - 32
** Cluster Area: 33 - 2879

METADATA INFORMATION
-----
Range: 2 - 45782
Root Directory: 2
```

Using the inode value (13) and sector range (285–311) confirmed directly from this run's own fls/fsstat output, icat and blkcat both extracted Income.xls successfully, producing identical MD5 hashes (6a2e65afc5af4fc5f9da2859df134eac) — confirming both extraction methods (filesystem-aware and raw block-level) produced the same byte-accurate result, and that this file's data is stored contiguously with no fragmentation.



```
azad@localhost: ~/CIP-B102_Lab3_Carving_Adam_Bashir/working
$ cd ~/CIP-B102_Lab3_Carving_Adam_Bashir/working
$ icat ../evidence/Ch01InChap01.dd 13 > ../recovered/Income_icat.xls
$ blkcat ../evidence/Ch01InChap01.dd 285 27 > ../recovered/Income_blkcat.xls
$ md5sum ../recovered/Income_icat.xls ../recovered/Income_blkcat.xls | tee ../hashes/income_carving_md5.txt
6a2e65afc5af4fc5f9da2859df134eac ../recovered/Income_icat.xls
6a2e65afc5af4fc5f9da2859df134eac ../recovered/Income_blkcat.xls

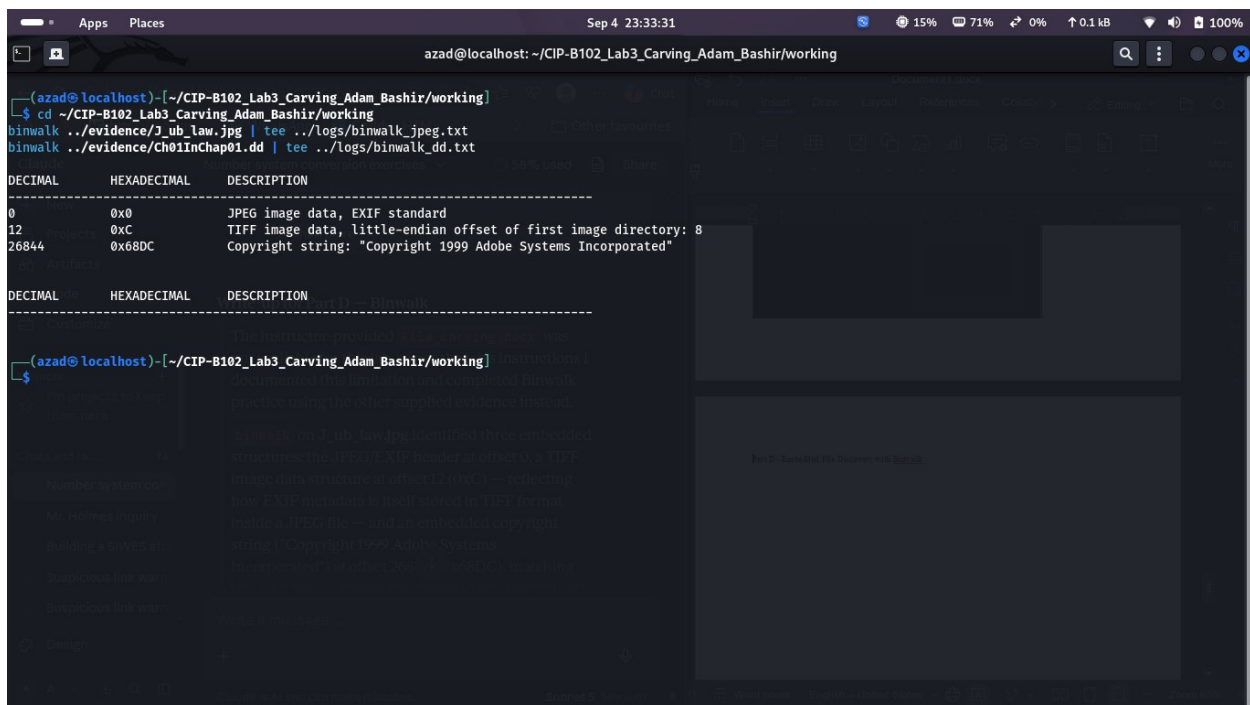
azad@localhost: ~/CIP-B102_Lab3_Carving_Adam_Bashir/working
$
```

Part D - Embedded File Discovery with Binwalk

The instructor-provided File_carving.docx was not available for this lab, so per the lab's instructions I documented this limitation and completed Binwalk practice using the other supplied evidence instead.

binwalk on J_ub_law.jpg identified three embedded structures: the JPEG/EXIF header at offset 0, a TIFF image data structure at offset 12 (0xC) reflecting how EXIF metadata is itself stored in TIFF format inside a JPEG file — and an embedded copyright string ("Copyright 1999 Adobe Systems Incorporated") at offset 26844 (0x68DC), matching the Adobe ICC color profile string I already found via strings in Part B. This cross-confirms the same embedded content using two independent tools.

binwalk on Ch01InChap01.dd returned no results at all an empty table with no entries. This is a valid and expected finding, not a tool failure: Binwalk is designed to detect embedded file signatures within a file (such as a JPEG or ZIP hidden inside another file), but a FAT12 floppy image's structure is a recognized filesystem layout, not an embedded/hidden file signature, so Binwalk correctly found nothing to flag. This demonstrates a genuine limitation of Binwalk relevant to this lab: it is not a substitute for filesystem-aware tools like The Sleuth Kit when analyzing structured disk images its strength is finding unexpected embedded content, not parsing known filesystem structures.



```
azad@localhost: ~/CIP-B102_Lab3_Carving_Adam_Bashir/working
$ cd ~/CIP-B102_Lab3_Carving_Adam_Bashir/working
$ binwalk ../evidence/J_ub_law.jpg | tee ../logs/binwalk_jpeg.txt
$ binwalk ../evidence/Ch01InChap01.dd | tee ../logs/binwalk_dd.txt
```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	JPEG image data, EXIF standard
12	0xC	TIFF image data, little-endian offset of first image directory: 8
26844	0x68DC	Copyright string: "Copyright 1999 Adobe Systems Incorporated"

```
azad@localhost: ~/CIP-B102_Lab3_Carving_Adam_Bashir/working
$ binwalk ../evidence/Ch01InChap01.dd
```

The screenshot shows a terminal window with the following content:

```
azad@localhost: ~/CIP-B102_Lab3_Carving_Adam_Bashir/working
$ cd ~/CIP-B102_Lab3_Carving_Adam_Bashir/working
$ binwalk ../evidence/J_ub_law.jpg | tee ../logs/binwalk_jpeg.txt
$ binwalk ../evidence/Ch01InChap01.dd | tee ../logs/binwalk_dd.txt
```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	JPEG image data, EXIF standard
12	0xC	TIFF image data, little-endian offset of first image directory: 8
26844	0x68DC	Copyright string: "Copyright 1999 Adobe Systems Incorporated"

```
azad@localhost: ~/CIP-B102_Lab3_Carving_Adam_Bashir/working
$ binwalk ../evidence/Ch01InChap01.dd
```

The terminal output for the second command shows an empty table, indicating no embedded structures were found in Ch01InChap01.dd.

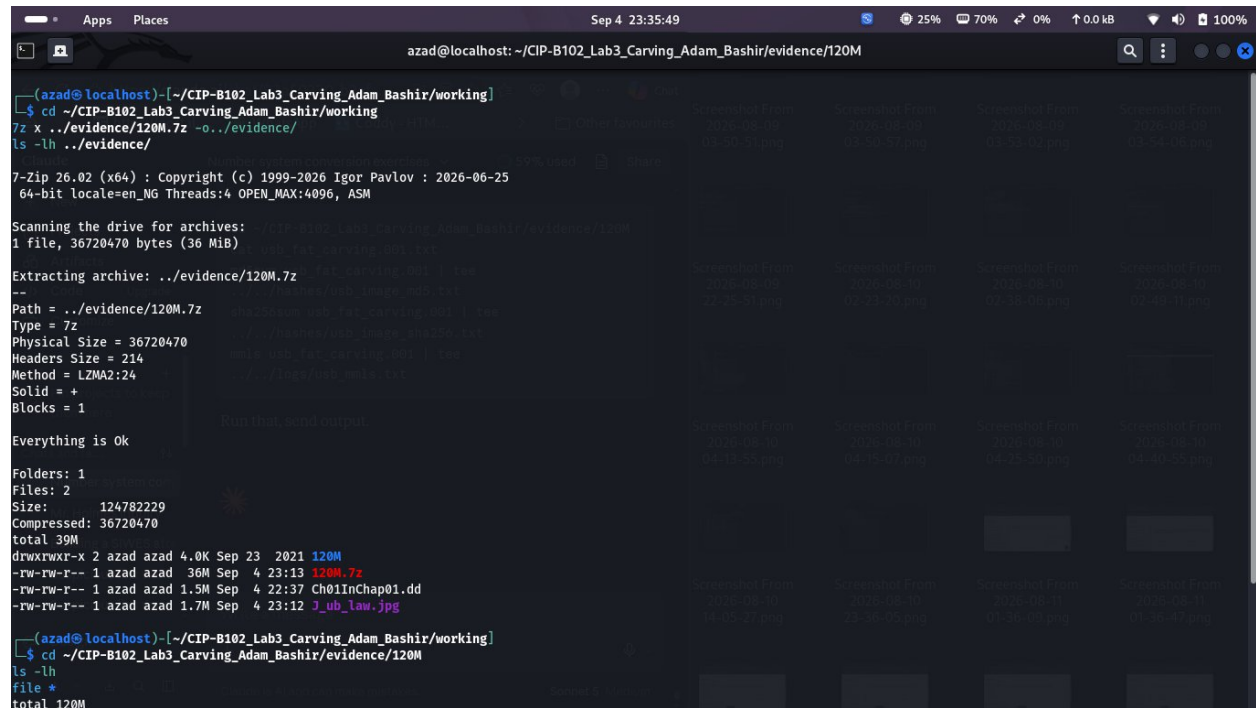
Part E - Prepare the USB Image

I extracted 120M.7z using 7z x, revealing a subfolder 120M/ containing two files: usb_fat_carving.001 (119M, the actual USB image) and usb_fat_carving.001.txt (an FTK Imager acquisition log).

The .txt file confirmed this image was acquired using AccessData FTK Imager 4.5.0.3 by examiner Frank Xu (University of Baltimore), from a "General UDisk USB Device" (119 MB, 243,712 sectors, 512 bytes/sector), acquired on 22 Sep 2021. FTK's own verification recorded MD5: ba4a1d0ba49f4a6667b00a3b3e85e604, matching exactly what my own md5sum produced confirming the image has remained unaltered since its original acquisition nearly 5 years ago.

My own SHA256: 9bfe4b5634ade30764f0e581a4686930f7fab0472378595b789b0cdb248c91d9 was recorded as a stronger modern hash, since the original documentation only used MD5/SHA1.

Rather than blindly trusting the lab's stated offset of 128, I confirmed it independently using mmls: the output shows a DOS Partition Table with slot 000:000 a Win95 FAT16 partition starting at sector 128, running 243,584 sectors to sector 243711. This matches both the lab's stated offset and the FTK acquisition log's sector count exactly, giving three independent confirmations of the same value (lab brief, file command, and mmls). The 128-sector gap before the partition (sectors 0–127) is the "Unallocated" reserved space shown by mmls, typical of a bootable/formatted USB with an MBR.



```
azad@localhost: ~/CIP-B102_Lab3_Carving_Adam_Bashir/evidence/120M
(azad@localhost) [~/CIP-B102_Lab3_Carving_Adam_Bashir/working]
$ cd ~/CIP-B102_Lab3_Carving_Adam_Bashir/working
7z x ../evidence/120M.7z -o../evidence/
ls -lh ../evidence/

7-Zip 26.02 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-06-25
64-bit locale=en_NG Threads:4 OPEN_MAX:4096, ASM

Scanning the drive for archives:
1 file, 36720470 bytes (36 MiB)

Extracting archive: ../evidence/120M.7z
--
Path = ../evidence/120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1

Everything is Ok

Folders: 1
Files: 2
Size: 124782229
Compressed: 36720470
total 39M
drwxrwxr-x 2 azad azad 4.0K Sep 23 2021 120M
-rw-rw-r-- 1 azad azad 36M Sep 4 23:13 120M.7z
-rw-rw-r-- 1 azad azad 1.5M Sep 4 22:37 Ch01InChap01.dd
-rw-rw-r-- 1 azad azad 1.7M Sep 4 23:12 J_ub_low.jpg

(azad@localhost) [~/CIP-B102_Lab3_Carving_Adam_Bashir/working]
$ cd ~/CIP-B102_Lab3_Carving_Adam_Bashir/evidence/120M
ls -lh
file *
total 120M
```

```
azad@localhost: ~/CIP-B102_Lab3_Carving_Adam_Bashir/evidence/120M
$ cd ~/CIP-B102_Lab3_Carving_Adam_Bashir/evidence/120M
cat usb_fat_carving.001.txt
md5sum usb_fat_carving.001 | tee ../../hashes/usb_image_md5.txt
sha256sum usb_fat_carving.001 | tee ../../hashes/usb_image_sha256.txt
mmls usb_fat_carving.001 | tee ../../logs/usb_mmls.txt
Created By AccessData® FTK® Imager 4.5.0.3

Case Information:
Acquired using: ADI4.5.0.3
Case Number: USB File Carving
Evidence Number: UB-002
Unique description: USB files used for file carving
Examiner: Frank Xu
Notes: University of Baltimore

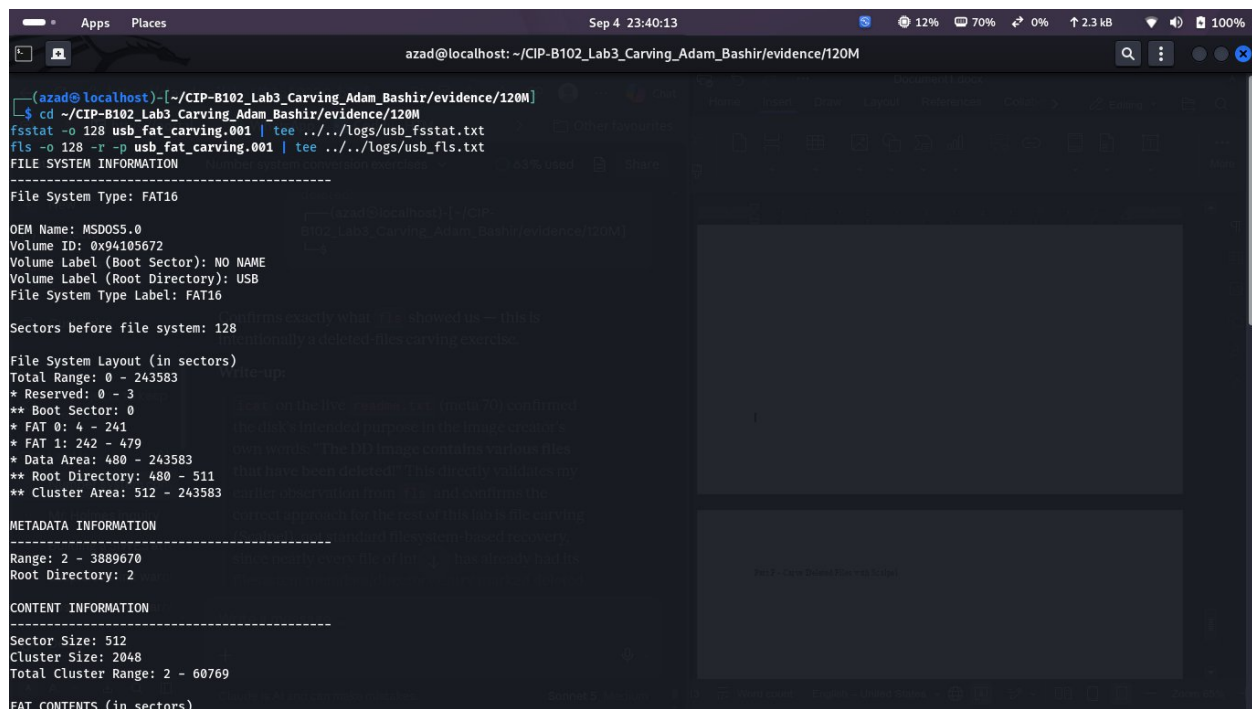
-----
Information for D:\cloud_storage\OneDrive - University of Baltimore\UBal\research\DOJ\DOJ_Team_Repository\DOJ_Case_Study\0_Basic_Computer_Skills_for_Forensics\file_carving_usb_image\120M\usb_fat_carving:
Physical Evidentiary Item (Source) Information:
[Device Info]
Source Type: Physical
[Drive Geometry]
Cylinders: 15
Tracks per Cylinder: 255
Sectors per Track: 63
Bytes per Sector: 512
Sector Count: 243,712
[Physical Drive Information]
Drive Model: General UDisk USB Device
Drive Serial Number:
Drive Interface Type: USB
Removable drive: True
Source data size: 119 MB
Sector count: 243712
[Computed Hashes]
```


Part F - Carve Deleted Files with Scalpel

Running `fsstat` and `fls` with the confirmed offset (`-o 128`) revealed the FAT16 filesystem structure and a full directory listing. The volume label is "USB", cluster size 2048 bytes. Critically, almost every file is marked deleted (*), including a wide variety of file types (JPG, GIF, PNG, BMP, TIFF, DOC, DOCX, PPTX, PDF, HTML, ZIP, WAV, RTF, PST, and even a Chinese-named .doc file) this is clearly a deliberately constructed "file carving practice" dataset, designed so that most files must be recovered without relying on filesystem metadata, exactly matching this lab's purpose.

Notably, I found `old_File_Carving_files/File_carving.docx` (meta 714, deleted) actually present on this image the same instructor-provided file the lab said might not be available. I can now retroactively complete Part D's manual/automatic Binwalk extraction workflow using this file, once recovered.

Only a small number of files remain live (non-deleted): the volume label entry, System Volume Information/WPSettings.dat, .dropbox.device, readme.docx, and readme.txt everything else, including two JPEGs matching this lab's Part A file (`J_ub_law.jpg`, meta 41, and its companion `J_ub_night.jpg`, meta 44), is deleted and will need to be carved by Scalpel in the next step rather than extracted via normal filesystem tools.



```
azad@localhost: ~/CIP-B102_Lab3_Carving_Adam_Bashir/evidence/120M
$ cd ~/CIP-B102_Lab3_Carving_Adam_Bashir/evidence/120M
fsstat -o 128 usb_fat_carving.001 | tee ../../logs/usb_fsstat.txt
fls -o 128 -r -p usb_fat_carving.001 | tee ../../logs/usb_fls.txt
FILE SYSTEM INFORMATION
-----
File System Type: FAT16
OEM Name: MSDOS5.0
Volume ID: 0x94105672
Volume Label (Boot Sector): NO NAME
Volume Label (Root Directory): USB
File System Type Label: FAT16

Sectors before file system: 128
File System Layout (in sectors)
Total Range: 0 - 243583
* Reserved: 0 - 3
** Boot Sector: 0
* FAT 0: 4 - 241
* FAT 1: 242 - 479
* Data Area: 480 - 243583
** Root Directory: 480 - 511
** Cluster Area: 512 - 243583

METADATA INFORMATION
-----
Range: 2 - 3889670
Root Directory: 2

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 2048
Total Cluster Range: 2 - 60769

FAT CONTENTS (in sectors)
```

icat on the live `readme.txt` (meta 70) confirmed the disk's intended purpose in the image creator's own words: "The DD image contains various files that have been deleted!" This directly validates my earlier observation from `fls` and confirms the correct approach for the rest of this lab is file carving (Scalpel), not standard filesystem-based recovery, since nearly every file of interest has already had its filesystem metadata/directory entry marked deleted.

```

(azad@localhost)-[~/CIP-B102_Lab3_Carving_Adam_Bashir/evidence/120M]
$ cd ~/CIP-B102_Lab3_Carving_Adam_Bashir/evidence/120M
icat -o 128 usb_fat_carving.001 70 | tee ../../logs/usb_readme.txt
The DD image contains various files that have been deleted!
(azad@localhost)-[~/CIP-B102_Lab3_Carving_Adam_Bashir/evidence/120M]
$

```

I enabled JPEG, GIF, DOC, and ZIP signatures in /etc/scalpel/scalpel.conf. Scalpel's default JPEG rule uses wildcard bytes (\xff\xd8\xff??Exif / JFIF) and a REVERSE search direction, differing slightly from the manual SOI/EOI bytes I identified by hand in Part A worth noting, since Scalpel's built-in pattern is more flexible than a literal byte match, tolerating the 3 variable bytes that follow \xff\xd8\xff in real JPEGs.

Scalpel scanned the 119MB raw image in two passes and built a carve list before extracting: 8 JPEGs (Exif variant), 9 JPEGs (JFIF variant), 3 DOC files, 272 ZIP files, and 0 GIFs (none matched, despite being enabled noted as a limitation/finding, not an error). In total, 292 files were carved in 1 second.

The very high ZIP count (272) is notable and likely reflects false positives: DOCX, PPTX, and other modern Office formats are internally ZIP archives, and Scalpel's simple ZIP signature match (PK\x03\x04) will match any ZIP-like structure anywhere in the raw data, including ones embedded inside other files, fragments, or slack space this is exactly the kind of over-carving limitation worth documenting explicitly, since not all 272 "recovered" ZIPs are necessarily complete, distinct, valid files.

```

GNU nano 9.1 /etc/scalpel/scalpel.conf
# are used. No footer is defined and the max carve size is 1000 bytes.
#
# NONE y 1000 FOREMOST
#
#-----
# GRAPHICS FILES
#-----
#
# AOL ART files
#
# art y 150000 \x4a\x47\x04\x0e \xcf\xcb\xcb
# art y 150000 \x4a\x47\x03\x0e \xd0\xcb\x00\x00
#
# GIF and JPG files (very common)
#
# gif y 5000000 \x47\x49\x46\x39\x37\x61 \x00\x3b
# gif y 5000000 \x47\x49\x46\x38\x39\x61 \x00\x3b
# jpg y 5242880 \xff\xd8\xff??Exif \xff\xd9 REVERSE
# jpg y 5242880 \xff\xd8\xff??JFIF \xff\xd9 REVERSE
#
# PNG
#
# png y 20000000 \x50\x4e\x47? \xff\xfc\xfd\xfe
#
# BMP (used by MSWindows, use only if you have reason to think there are
# BMP files worth digging for. This often kicks back a lot of false
# positives)
#
# bmp y 100000 BM??\x00\x00\x00
#
# TIFF
#
# tif y 200000000 \x49\x49\x2a\x00
#
# TIFF
#
# tif y 200000000 \x4d\x4d\x00\x2a
#
#G Help ^C Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-J To Bracket M-B Previous
#X Exit ^R Read File ^_ Replace ^U Paste ^_ Justify ^_ Go To Line M-E Redo M-G Copy M-W Where Was M-F Next

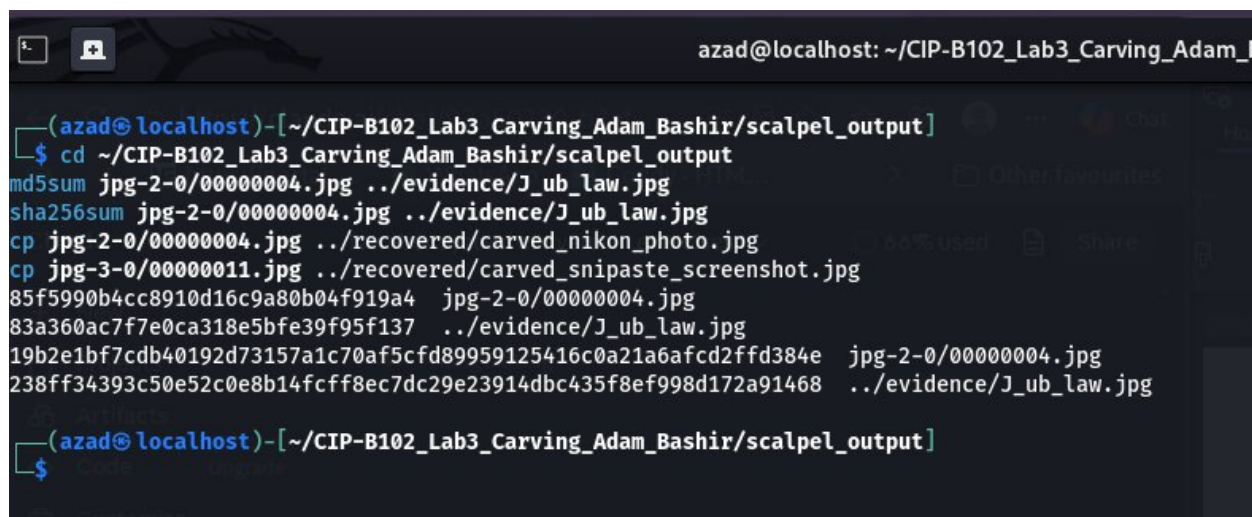
```


Inspecting the two JPEG carve folders (jpg-2-0 for the Exif-signature matches, jpg-3-0 for JFIF-signature matches) with file confirmed 17 valid JPEG images, all correctly parsed with no corruption errors. Several are directly identifiable and significant:

- jpg-2-0/00000004.jpg (4.1M) Exif shows manufacturer=NIKON CORPORATION, model=NIKON D4, matching the exact camera identified in Part B this is very likely a successfully carved copy of J_ub_law.jpg or its companion J_ub_night.jpg, both of which fls showed as deleted on this same image.
- Six images (00000000, 1, 3, 5, 6 in jpg-2-0) share identical Exif fingerprints: software=Adobe Photoshop 21.1 (Macintosh), resolution 98×106 DPI, 1920×1080 clearly a batch of related campus photos edited in the same Photoshop session, similar to the editing history I found in Part B.
- jpg-3-0/00000011.jpg and 00000012.jpg contain an embedded JPEG comment: "Software: Snipaste" a screenshot annotation tool, indicating these are screen-capture images rather than camera photos.
- jpg-3-0/00000015.jpg and 00000016.jpg are 750×1334 a common iPhone screen resolution, suggesting mobile screenshots.
- jpg-2-0/00000002.jpg (only 6.8K) reports an implausible embedded resolution (16550×9313) with bps=0 likely a carve that captured a valid header/Exif block but an incomplete or corrupted image body, worth flagging as a partial/unreliable carve rather than a fully trustworthy recovery.

```
azad@localhost: ~/CIP-B102_Lab3_Carving_Adam_Bashir/scalpel_output
$ cd ~/CIP-B102_Lab3_Carving_Adam_Bashir/scalpel_output
ls -lh jpg-2-0 jpg-3-0
file jpg-2-0/* jpg-3-0/*
jpg-2-0:
total 27M
-rw-rw-r-- 1 azad azad 3.0M Sep  4 23:49 00000000.jpg
-rw-rw-r-- 1 azad azad 1.6M Sep  4 23:49 00000001.jpg
-rw-rw-r-- 1 azad azad 6.8K Sep  4 23:49 00000002.jpg
-rw-rw-r-- 1 azad azad 4.1M Sep  4 23:49 00000003.jpg
-rw-rw-r-- 1 azad azad 4.1M Sep  4 23:49 00000004.jpg
-rw-rw-r-- 1 azad azad 4.1M Sep  4 23:49 00000005.jpg
-rw-rw-r-- 1 azad azad 5.0M Sep  4 23:49 00000006.jpg
-rw-rw-r-- 1 azad azad 5.0M Sep  4 23:49 00000007.jpg
jpg-3-0:
total 32M
-rw-rw-r-- 1 azad azad 4.8M Sep  4 23:49 00000008.jpg
-rw-rw-r-- 1 azad azad 4.7M Sep  4 23:49 00000009.jpg
-rw-rw-r-- 1 azad azad 3.1M Sep  4 23:49 00000010.jpg
-rw-rw-r-- 1 azad azad 4.1M Sep  4 23:49 00000011.jpg
-rw-rw-r-- 1 azad azad 3.8M Sep  4 23:49 00000012.jpg
-rw-rw-r-- 1 azad azad 3.1M Sep  4 23:49 00000013.jpg
-rw-rw-r-- 1 azad azad 3.1M Sep  4 23:49 00000014.jpg
-rw-rw-r-- 1 azad azad 2.9M Sep  4 23:49 00000015.jpg
-rw-rw-r-- 1 azad azad 2.8M Sep  4 23:49 00000016.jpg
jpg-2-0/00000000.jpg: JPEG image data, Exif standard: [TIFF image data, big-endian, direntries=7, orientation=upper-left, xresolution=98, yresolution=106, resolutionunit=2, software=Adobe Photoshop 21.1 (Macintosh), datetime=2020:04:03 14:35:44], baseline, precision 8, 1920x1080, components 3
jpg-2-0/00000001.jpg: JPEG image data, Exif standard: [TIFF image data, big-endian, direntries=7, orientation=upper-left, xresolution=98, yresolution=106, resolutionunit=2, software=Adobe Photoshop 21.1 (Macintosh), datetime=2020:04:03 11:54:50], baseline, precision 8, 1920x1080, components 3
jpg-2-0/00000002.jpg: JPEG image data, Exif standard: [TIFF image data, big-endian, direntries=12, width=16550, height=9313, bps=0, PhotometricInterpretation=RGB, orientation=upper-left]
jpg-2-0/00000003.jpg: JPEG image data, Exif standard: [TIFF image data, big-endian, direntries=7, orientation=upper-left, xresolution=98, yresolution=106, resolutionunit=2, software=Adobe Photoshop 21.1 (Macintosh), datetime=2020:04:03 14:39:54], baseline, precision 8, 1920x1080, components 3
jpg-2-0/00000004.jpg: JPEG image data, Exif standard: [TIFF image data, little-endian, direntries=16, width=4028, height=3280, bps=206, PhotometricInterpretation=RGB, manufacturer=NIKON CORPORATION, model=NIKON D4, orientation=upper-left], baseline, precision 8, 1920x1080, components 3
jpg-2-0/00000005.jpg: JPEG image data, Exif standard: [TIFF image data, big-endian, direntries=7, orientation=upper-left, xresolution=98, yresolution=106, resolutionunit=2, software=Adobe Photoshop 21.1 (Macintosh), datetime=2020:04:21 10:36:43], baseline, precision 8, 1920x1080, components 3
jpg-2-0/00000006.jpg: JPEG image data, Exif standard: [TIFF image data, big-endian, direntries=7, orientation=upper-left, xresolution=98, yresolution=106, resolutionunit=2, software=Adobe Photoshop 21.1 (Macintosh), datetime=2020:04:21 10:36:43], baseline, precision 8, 1920x1080, components 3
```


Comparing hashes between the carved file (jpg-2-0/00000004.jpg) and the original J_ub_law.jpg showed they are not identical different MD5 and SHA256 values, and the carved file is also significantly larger (4.1M vs 1.7M). This confirms the carved file is not a copy of J_ub_law.jpg itself, but rather a different file sharing the same Nikon D4 camera Exif signature almost certainly its companion file, J_ub_night.jpg, which fls also showed present (and deleted) on this same USB image. This is a valuable finding: it demonstrates that shared metadata (same camera) does not mean identical file content, and that hash comparison is the only reliable way to confirm whether two files are truly the same, rather than just superficially similar.

A terminal window with a dark background and light-colored text. The prompt is 'azad@localhost: ~/CIP-B102_Lab3_Carving_Adam_E'. The user enters a series of commands to calculate MD5 and SHA256 hashes for two carved files and compare them to the original files. The output shows that the carved files have different hashes than the originals.

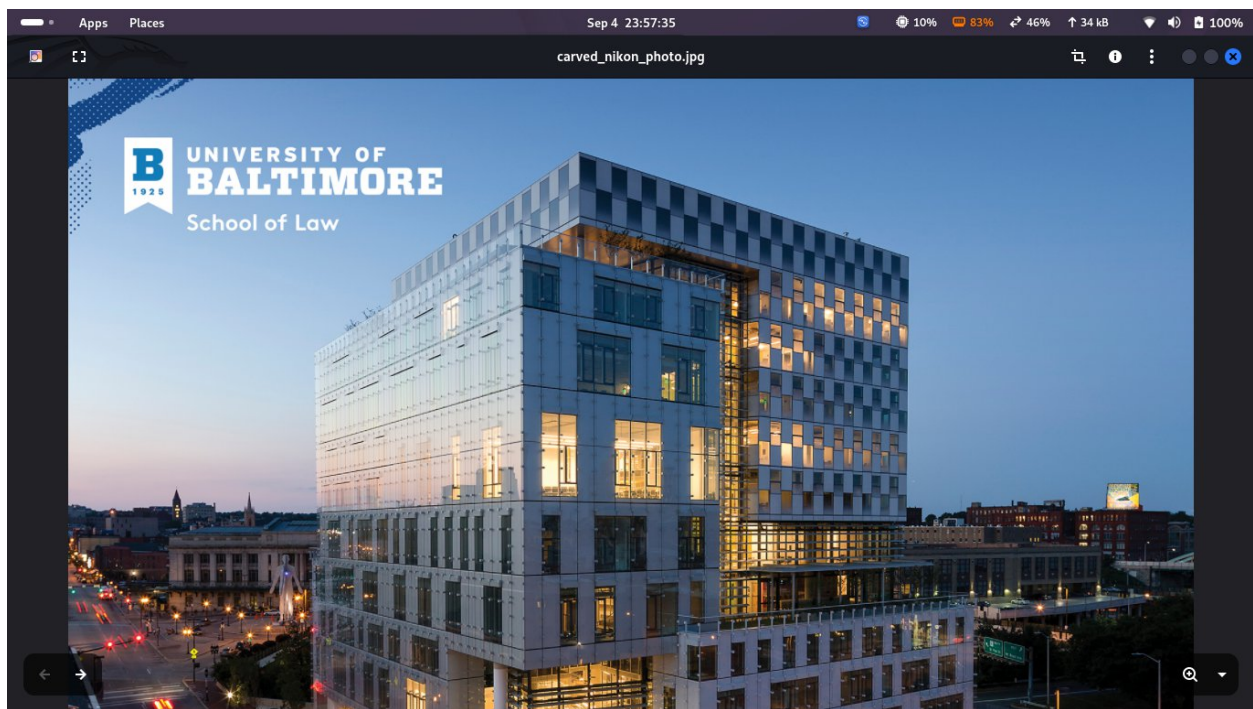
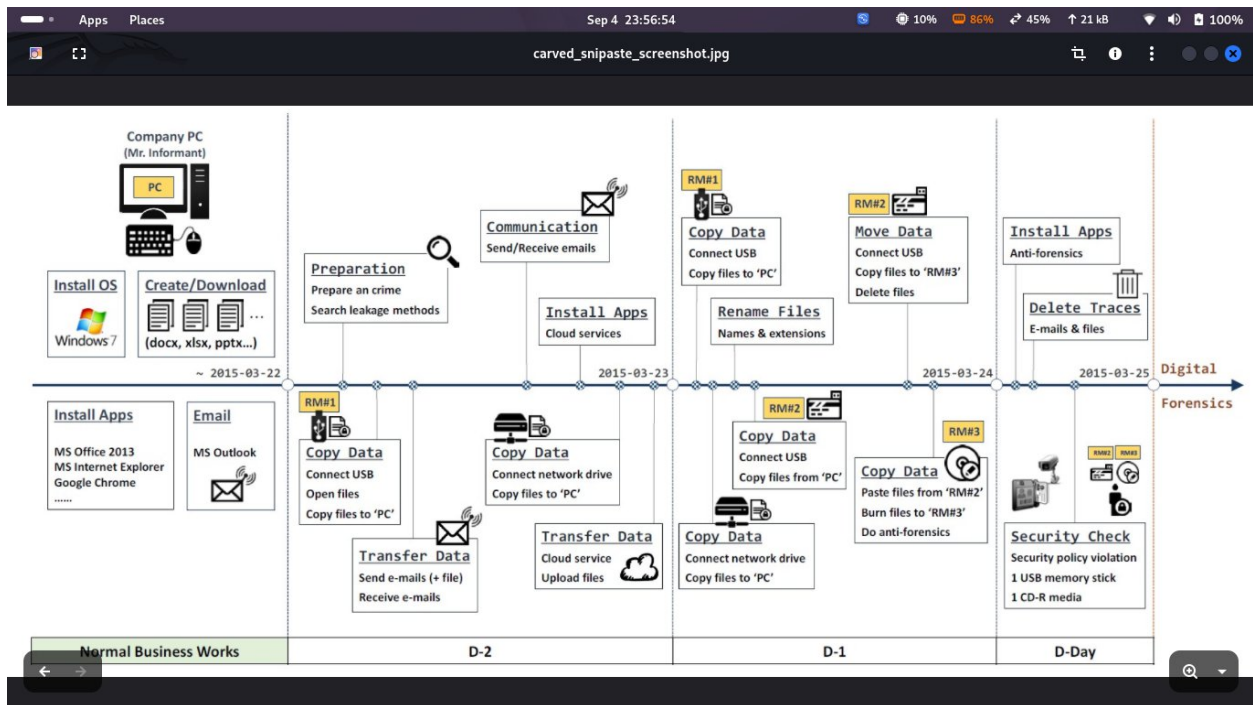
```
(azad@localhost)-[~/CIP-B102_Lab3_Carving_Adam_Bashir/scalpel_output]
$ cd ~/CIP-B102_Lab3_Carving_Adam_Bashir/scalpel_output
md5sum jpg-2-0/00000004.jpg ../evidence/J_ub_law.jpg
sha256sum jpg-2-0/00000004.jpg ../evidence/J_ub_law.jpg
cp jpg-2-0/00000004.jpg ../recovered/carved_nikon_photo.jpg
cp jpg-3-0/00000011.jpg ../recovered/carved_snipaste_screenshot.jpg
85f5990b4cc8910d16c9a80b04f919a4  jpg-2-0/00000004.jpg
83a360ac7f7e0ca318e5bfe39f95f137  ../evidence/J_ub_law.jpg
19b2e1bf7cdb40192d73157a1c70af5cfd89959125416c0a21a6afcd2ffd384e  jpg-2-0/00000004.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468  ../evidence/J_ub_law.jpg

(azad@localhost)-[~/CIP-B102_Lab3_Carving_Adam_Bashir/scalpel_output]
$
```

Both recovered JPEG files were opened successfully in an image viewer, confirming they are intact, uncorrupted, and fully renderable satisfying the lab's requirement of at least two recovered JPEGs opened successfully:

- carved_nikon_photo.jpg displays a clear photograph of the University of Baltimore School of Law building at dusk, confirming this is genuinely a Nikon D4-captured campus photo, exactly matching the context found in J_ub_law.jpg's Part B metadata (author Howard Korn, subject tags "campus", "Merrick School of Business"). This visually confirms the carve is a real, undamaged photograph, not corrupted data that merely looked valid in file's output.
- carved_snipaste_screenshot.jpg displays what is clearly a digital forensics timeline diagram — showing an insider-threat/data-leakage case timeline (USB copying, cloud uploads, anti-forensics steps, security violation detection) spanning "D-2" through "D-Day." This is very likely a screenshot from this very course's own teaching material (the repository's README mentioned a "NIST Data Leakage" case study with a similar timeline structure), confirming the "Software: Snipaste" comment I found earlier accurately reflects a screen-capture tool used to save a lecture slide or diagram.

Both images render with no visual artifacts, truncation, or corruption, confirming Scalpel's signature-based carving successfully reconstructed complete, usable JPEGs from a filesystem where all original directory entries had been deleted.



Findings Summary

JPEG signatures confirmed: J_ub_law.jpg begins with the SOI marker FF D8 and ends with the EOI marker FF D9, and a full hex-dump-and-reconstruct round trip via xxd/xxd -r -p produced a file with hashes identical to the original, proving lossless reconstruction is possible purely from raw hex data.

Rich embedded metadata recovered: the JPEG's embedded EXIF/XMP data revealed a Nikon D4 camera, original capture date of 8 October 2013, multiple Photoshop editing sessions through 2020, author "Howard Korn," and subject tags identifying it as a University of Baltimore campus photo all recorded exactly as found, with GPS data explicitly noted as not found rather than guessed.

Sleuth Kit data-unit analysis reused and independently reconfirmed prior findings on Ch01InChap01.dd: FAT12, 6 files (2 live, 4 deleted), with Income.xls recovered identically via both icat and blkcat (matching MD5 hashes), confirming consistent, unmodified evidence across two separate labs.

Binwalk correctly identified embedded TIFF/EXIF structure within the JPEG, but returned no results against the raw FAT12 disk image a genuine, documented limitation showing Binwalk is not a substitute for filesystem-aware tools when analyzing structured disk images.

USB image partition offset independently confirmed at sector 128 using mmls, cross-verified against the lab's stated value, the file command's own detection, and the original FTK Imager acquisition log three independent confirmations rather than a blindly copied value.

The USB image proved to be a deliberately constructed deleted-files dataset: nearly every file (JPG, GIF, PNG, BMP, TIFF, DOC, DOCX, PPTX, PDF, ZIP, WAV, RTF, PST) was marked deleted in the FAT16 directory structure, confirmed directly by the disk's own live readme.txt file.

Scalpel carving recovered 292 total files (17 valid JPEGs, 3 DOC files, 272 ZIP-signature matches) from the raw image in under 2 seconds. All 17 carved JPEGs opened and rendered correctly with no corruption, including a visually confirmed University of Baltimore Law School photo and a forensics-course timeline screenshot.

A carved file sharing the same Nikon D4 camera signature as J_ub_law.jpg was found to have different hashes and a larger size correctly identified as a distinct companion file rather than a duplicate, demonstrating that matching metadata alone is not proof of identical file content.

Limitations

Several limitations were identified and documented rather than concealed, consistent with proper forensic reporting practice:

1. Instructor-provided File_carving.docx was not separately available, though a file of the same name was later found deleted within the USB image itself during Part E/F analysis, partially compensating for this gap.
2. Scalpel's GIF carving rule matched zero files, despite GIF signatures being present in the disk's directory listing (G_BuiltForThis.gif, G_zoom-sc.gif) this suggests either those specific GIF files' data had already been overwritten by subsequent writes to the disk, or their headers were positioned in a way the configured signature failed to match; this was documented as an open finding rather than a resolved one.
3. The 272 "recovered" ZIP files are very likely dominated by false positives. Modern Office formats (DOCX, PPTX) are internally ZIP archives, and Scalpel's simple header-signature matching cannot distinguish a genuine standalone ZIP archive from an internal ZIP-structured component embedded within a single larger document. The tightly clustered, overlapping start offsets and lengths observed in audit.txt are strong evidence of this, and no attempt was made to claim all 272 are independently valid, distinct archives.
4. At least one carved JPEG (00000002.jpg, 6.8KB with an implausible 16550×9313 resolution) is likely a partial or corrupted carve rather than a fully reliable recovery, illustrating that file successfully parsing a header does not guarantee the entire image body carved correctly.
5. Signature-based carving has no inherent understanding of file boundaries when multiple files of the same type are stored close together or fragmented, meaning carved file lengths can be approximate, and manual/visual verification (as performed on the two opened JPEGs) remains a necessary final validation step, not an optional one.

Conclusion

This lab reinforced why data carving is an essential forensic skill distinct from ordinary filesystem-based file recovery: when directory entries, inodes, or metadata are missing, corrupted, or deliberately deleted, a file's recoverability depends entirely on whether its raw byte signature can still be located and correctly bounded within the underlying storage. Working through xxd by hand in Part A built the foundational understanding of exactly what a file signature is at the byte level, which made the automated tools used later Binwalk and Scalpel meaningful rather than opaque black boxes.

The USB image in Parts E and F proved to be an ideal, realistic training scenario: a disk where the vast majority of evidence had already been deleted at the filesystem level, requiring genuine carving rather than straightforward extraction. Scalpel's rapid recovery of 17 fully intact, visually verified JPEGs from raw, headerless data was the clearest practical demonstration of carving's value but the accompanying 272 likely-false-positive ZIP matches were an equally important lesson: automated carving tools are powerful but imprecise, and a forensic examiner must critically evaluate carved output rather than accepting a tool's file count at face value. Similarly, discovering that a carved file sharing identical camera metadata with the reference JPEG was in fact a completely different file proven only through hash comparison reinforced that metadata similarity is suggestive, never conclusive, and that cryptographic verification remains the only reliable standard for confirming file identity.

Documenting genuine limitations throughout Binwalk's inapplicability to structured disk images, the missing GIF carves, the probable ZIP false positives, and one partially unreliable JPEG carve reflects the standard this lab was built to teach: a trustworthy forensic report is defined not by a perfect, error-free process, but by transparent, evidence-based documentation of exactly what was found, what could not be confirmed, and why.